Demography Predicts Genetic Effective Size in a Desert Stream Fish Community

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ABSTRACT: Patterns of genetic diversity and effective size should be predicted by life history traits (intrinsic), landscape properties (extrinsic), and population dynamics. Theoretical models portray complicated relationships among population subdivision, rates of extirpation and recolonization, and metapopulation genetic effective size (*metaN*$_e$) but make simplifying assumptions about effects of intrinsic and extrinsic factors. Using previously published data sets, we compared estimates of genetic effective size to demographic time-series data gathered for nine dominant species in a desert stream fish community. These species occupy a common desert stream network and experience the same disturbance regime but differ in abundance, distribution, and life history traits that should affect reproduction, dispersal, local persistence, and genetic diversity patterns. Measures of genetic effective size were positively related to network-wide abundance and mean adult density across species and were negatively related to extirpation probability. Comparative data supported the theoretical prediction that population subdivision decreases metapopulation genetic effective size relative to panmictic populations of the same size. Estimates of *metaN*$_e$ reflected differences in intrinsic traits and population dynamics across species measured over ecological timescales. This comparative study exemplifies why ecological differences are important considerations when seeking to preserve genetic diversity.

Keywords: metapopulation genetic effective size, gene flow, life history, disturbance, community genetics, river network.

Introduction

Unraveling mechanisms that drive intraspecific genetic diversity and influence evolutionary trajectories is a long-standing focus of evolutionary ecology, with increasing applications toward biodiversity conservation. Whereas relationships between abundance and genetic diversity have received much attention, few studies using vertebrates have investigated the relationship between genetic diversity and short-term demographic turnover. Theoretical and empirical work shows that intra- and interpopulation genetic diversity is modulated by emergent properties of a species' population and its environment that are classified into three broad mechanistic categories: intrinsic traits, extrinsic factors, and (meta)population dynamics. Over ecological timescales, these mechanisms should have measurable impacts on genetic drift and gene flow, two evolutionary forces responsible for contemporary standing levels of genetic diversity.

The first category encompasses species intrinsic traits as defined by the fundamental niche (sensu Hutchinson 1957) that dictate the conditions under which a species can survive and reproduce. Specific intrinsic traits, such as mode of reproduction (e.g., sexual vs. clonal), generation time, and vagility, constrain the rates at which gene flow and genetic drift act on populations. Comparative investigations confirm that levels of genetic differentiation vary predictably across a broad assortment of taxa with different traits. For example, the dispersal mode of plants (Givnish 2010; Meirmans et al. 2011) and aquatic macroinvertebrates (Phillipsen et al. 2015) as well as the habitat preference of birds (Burney and Brumfield 2009) and freshwater fishes (Tibbets and Dowling 1996) are intrinsic traits associated with genetic differentiation. Yet the relationship between intrinsic traits and genetic patterns is not always clear because of unanticipated interactions with extrinsic factors or historical contingency (e.g., Galarza et al. 2009; Faurby and Barber 2012).

Extrinsic factors shape genetic patterns through processes that constrain gene flow, allowing genetic drift to

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act independently among groups within a population (Slatkin 1993). Heterogeneous spatial distribution of resources, network configuration of dispersal corridors, and other factors that promote isolation across a network of habitat patches cause demographic population subdivision. Extrinsic constraints on gene flow manifest across species regardless of intrinsic traits (e.g., Galarza et al. 2009; Riginos et al. 2011; Treml et al. 2015), which is why comparative studies using codistributed species within the same network of habitat patches are necessary to "control" for shared influences of extrinsic factors.

Demographic population subdivision has two important consequences for neutral genetic variation. First, as gene flow decreases among population segments (herein called subpopulations), differentiation of allelic frequencies increases, measured as F_{ST}. Wright (1943) formulated the equation $F_{ST} \approx 1/(1 + 4N_e m)$ to describe the equilibrium relationship between genetic differentiation and genetically effective number of migrants ($N_e m$) that have equal probability of migrating to any subpopulation in an equally connected network of discrete habitats (i.e., the island model). Second, the probability of an allele going to fixation across all subpopulations decreases monotonically as F_{ST} increases, thereby countering the homogenizing effects of genetic drift at the metapopulation scale. Rearranging the previous equation, the equilibrium effective size of the metapopulation (hereafter referred to as $metaN_e$; table 1) can be derived, $metaN_{eIsl} = nN_e/(1 - F_{ST})$, where n is the number of subpopulations assumed to have equal N_e and m among them (Wang and Caballero 1999). Others have generalized these equations to apply under less strict equilibrium assumptions (see Wang and Caballero 1999).

Ultimately, patterns of genetic variation emerge from interactions between species' intrinsic traits and extrinsic factors that are realized through metapopulation processes (e.g., source-sink or extirpation-colonization dynamics). Following Levins (1969) and Hanski (1998), we define a metapopulation as a group of interconnected subpopulations that occupy discrete habitat patches. Occupancy fluctuates between extirpations and colonization of unoccupied patches. Extrinsic factors, such as heterogeneity in habitat size, quality, and disturbance regime, affect the rate of extirpations. Intrinsic traits, such as intrinsic growth rate and vagility, influence a species' resistance to extirpation (persisting in a patch, even if at low densities) or resilience to extirpations (how quickly a patch is recolonized). Metapopulations can be at a demographic equilibrium between extirpation and colonization events, but variation in subpopulation size and age (i.e., time since colonization) prevent gene flow and genetic drift from reaching equilibrium in the classic island model of population structure (Wright 1943). Because equilibrium assumptions are unlikely to be met, it is inadvisable to estimate the product of N_e and m from F_{ST} (Whitlock and McCauley 1999).

Theoretical work has shown that variance in subpopulation size and productivity, complex spatial arrangements, asymmetrical gene flow, and extirpation-colonization dynamics can increase or decrease F_{ST} among subpopulations relative to the island model (for reviews, see Wang and Caballero 1999; Whitlock 2004). The effect on overall genetic differentiation is primarily driven by the source of colonizers (Whitlock and McCauley 1990). However, a consequence of these conditions is to enhance variance in reproductive output among subpopulations, which will

Table 1: Symbol notation and definitions used in this study

	Definition
F_{ST}	Genetic differentiation in allele frequencies among all subpopulations
N_{bi}	Genetic effective number of breeders for subpopulation i
$\sum N_{bi}$	Sum of subpopulation-specific number of breeders
N_{ei}	Genetic effective size per generation for subpopulation i
$\sum N_{ei}$	Sum of subpopulation-specific effective sizes
m	Migration rate per generation
n	Number of subpopulations
$metaN_e$	Metapopulation genetic effective size for a group of interconnected subpopulations; subscripted suffixes apply to specific models
P_e	Extirpation probability per generation
C	Index of commonness
A	Index of asymmetric gene flow calculated as the absolute value of the summed differences in unidirectional migration rates
A_r	Allelic richness calculated for each subpopulation, accounts for sample size differences
ϕ	Probability of common origin from source populations for individuals recolonizing unoccupied habitats
k	Number of individuals recolonizing unoccupied habitats

decrease $metaN_e$ relative to the sum of all local effective sizes ($\sum N_{ei}$, where N_{ei} is the effective population size of subpopulation *i*). Theoretically, $metaN_e$ will be greater than $\sum N_{ei}$ only if there is no variability in reproductive output across the metapopulation (Whitlock 2004).

Although theory predicts that $metaN_e$ will be less than $\sum N_{ei}$ under biologically realistic conditions, there have been relatively few empirical investigations to test these predictions. Studies in salmonids provide support for population subdivision decreasing $metaN_e$ relative to $\sum N_{ei}$ (Fraser et al. 2007; Kuparinen et al. 2010; Palstra and Ruzzante 2011). Morrissey and de Kerckhove (2009) demonstrated through simulations that gene flow asymmetry in dendritic networks could reduce the probability of an allele going to fixation over the whole metapopulation, thus increasing $metaN_e$. Yet salmonids experiencing asymmetrical gene flow exhibited decreased $metaN_e$ relative to $\sum N_{ei}$ (Kuparinen et al. 2010; Gomez-Uchida et al. 2013). Temporal instability in the form of extirpations and recolonizations reduced $metaN_e$ in a metapopulation of mycophagous beetles (Ingvarsson et al. 1997). Elsewhere, metapopulations of ephemeral midge species experiencing repeated local extirpations had reduced genetic diversity at the local scale compared with species with temporally stable populations, but higher regional genetic diversity (and presumably higher $metaN_e$) was maintained through high gene flow because of the nomadic dispersal trait of the ephemeral species (Berendonk et al. 2009). Using simulated populations, Ray (2001) emphasized that the spatial scale of local extirpations relative to population subdivision was important for determining the loss of genetic diversity. Except for Berendonk et al. (2009), whose study included six species, previous research has focused on three or fewer species. These few and contrasting studies highlight the complexities involving intrinsic traits, metapopulation dynamics, and $metaN_e$ and the need for additional empirical studies across multiple species with different intrinsic traits that share a common environment. Studies of communities of organisms are an appropriate means to examine such interactions (Wares 2002).

Previously, we found that species-specific patterns of genetic differentiation (i.e., F_{ST}) among subpopulations for nine codistributed fish species occupying a dynamic stream network could be explained by aspects of the river network, intrinsic traits, and genetic drift (Pilger et al. 2017). Using the same genotypic data, here we focus on the causes of genetic drift within and among subpopulations—specifically, how intrinsic properties may either mitigate or exacerbate effects of genetic drift in metapopulations experiencing recurring extirpations. The dynamic nature of the system prevents species from conforming to migration-drift equilibrium; therefore, we first asked whether variation in standing levels of genetic diversity among species could be explained by variation in relative abundance, occupancy, adult density, and extirpation probabilities among species. As others have documented relationships between abundance and neutral genetic diversity (e.g., McCusker and Bentzen 2010; Osborne et al. 2010; Hague and Routman 2016), we predicted that genetic diversity would be positively associated with relative abundance, occupancy, and adult density. We predicted extirpation probabilities would be negatively associated with genetic diversity, consistent with the notion that increasing extirpation rates reduce $metaN_e$ (Whitlock and Barton 1997; Whitlock 2004).

Second, we asked how intrinsic traits and metapopulation dynamics might interact on a shared landscape to influence $metaN_e$. We identified trait differences across species that likely resulted in differences in abundance and gene flow, ultimately shaping $metaN_e$. Because some species in a community are more closely related than others, we acknowledge that phylogenetic inertia could play a role in explaining outcomes of metapopulation processes over ecological time.

Last, we employed three $metaN_e$ models that have different assumptions related to migration-drift equilibrium (table 2) to ask what these models can reveal about the effects of genetic drift occurring at the metapopulation scale. Estimates from each $metaN_e$ model were compared with $\sum N_{ei}$, the expected $metaN_e$ if subpopulations formed a single panmictic population (i.e., complete random mating) by calculating the ratio of $metaN_e$ to $\sum N_{ei}$. The pooled model (Palstra and Ruzzante 2011) is the least restrictive, as it makes no assumption of equilibrium. As a practical model, samples are pooled across all subpopulations to estimate $metaN_{ePool}$ using the same estimation method (e.g., Hill's [1981] linkage disequilibrium method) as used to estimate N_e for each subpopulation. Because pooling samples to estimate $metaN_e$ is sensitive to the amount of differentiation among samples (Palstra and Ruzzante 2011; Turner et al. 2015), the $metaN_{ePool}:\sum N_{ei}$ ratio will be 1.0 only if $F_{ST} = 0$. We predicted $metaN_{ePool}:\sum N_{ei}$ to decrease as global F_{ST} increased across species. The theoretical island (Wright 1931) and extirpation (Whitlock and Barton 1997) models are based on ideal populations with restrictive assumptions, namely, that all subpopulations have the same constant number of individuals, that all subpopulations contribute equally to the next generation, and that migration is random among subpopulations. Whereas the island model assumes that migration and genetic drift are in equilibrium, the extirpation model relaxes this assumption and instead assumes that all subpopulations have an equal probability of becoming extirpated. Violations of these assumptions are expected to reduce $metaN_e:\sum N_{ei}$ for both models. Therefore, we predicted that increasing asymmetric migration or greater variation among local subpopulation size would correspond to decreased $metaN_e:\sum N_{ei}$ ratios across species. For the extirpation

Table 2: $metaN_e$ models used on nine species of desert fishes

Model	Parameterization	Assumptions	Predictions
Pooled (Palstra and Ruzzante 2011)	$metaN_{ePool} = N_e$ for all samples combined	Panmixia, $F_{ST} = 0$	$metaN_e < \sum N_{ei}$ $metaN_e : \sum N_{ei}$ ratio decreases with increasing F_{ST} (Palstra and Ruzzante 2011; Turner et al. 2015)
Island (Wright 1943)	$metaN_{eIsl} = \frac{\sum_{i=1}^{S} N_{ei}}{1 - F_{ST}}$	Equilibrium conditions leading to $F_{ST} = \frac{1}{4Nm + 1}$	$metaN_e = \sum N_{ei}$, if $F_{ST} = 0$
	N_{ei} = local effective size F_{ST} = global F_{ST}	Equal N_{ei} for all subpopulations, no extirpation, random mating, subpopulations receive equal migrants drawn randomly from migrant pool	$metaN_e > \sum N_{ei}$, for $F_{ST} > 0$ $metaN_e : \sum N_{ei}$ ratio increases with increasing F_{ST} (Wright 1943)
Extirpation-recolonization (Whitlock and Barton 1997)	$metaN_{eEx} = \frac{n}{4(m + e)F_{ST}}$	Equal N_{ei} for all subpopulations, symmetric migration, colonists have equal probability of being from any extant subpopulations	$metaN_e < \sum N_{ei}$
	n = number of subpopulations		$metaN_e : \sum N_{ei}$ ratio decreases with increased gene flow asymmetry
	m = migration rate per generation e = extirpation rate per generation		$metaN_e : \sum N_{ei}$ ratio decreases with increasing F_{ST} (Whitlock and Barton 1997)

Note: Parameter estimates used for calculations of $metaN_{eIsl}$ and $metaN_{eEx}$ are provided in table 3. Assumptions of each model are the conditions that if met would result in $metaN_e = \sum N_{ei}$. The predictions column provides the expectation of how the ratio of $metaN_e$ to $\sum N_{ei}$ will respond when the assumptions are not met.

model specifically, Whitlock and Barton (1997) suggest that differentiation should decrease $metaN_{eEx}$; therefore, we also predicted that increasing F_{ST} across species would coincide with a reduction in $metaN_{eEx} : \sum N_{ei}$.

Methods

Study System

Much of the Colorado River Basin has been extensively engineered for water storage and conveyance, but the study area in the upper Gila River region (a major tributary of the lower Colorado River) remains largely unmodified and retains a natural flow regime (Propst et al. 2008). Arising in the Mogollon Mountains in southwestern New Mexico, the upper Gila River drainage transitions from small, shallow, high-gradient tributaries to a large, deep, low-gradient mainstem. High-elevation streams in the network are narrow, with steep hill slopes and dense riparian vegetation (Brown 1982) that limit instream primary production (Whitney et al. 2014). Lower-elevation mainstem reaches consist of braided channels with high floodplain connectivity (Soles 2016). Like other lotic systems in the arid US Southwest, the Gila River has high inter- and intra-annual variation in river discharge associated with winter–spring snowmelt (December–May) and rainfall from the North American monsoon (July–September; Gutzler 2013; Pascolini-Campbella et al. 2015). Primary disturbances include supraseasonal droughts, extreme flooding, and large wildfires (Propst et al. 2008; Gido et al. 2019). Individual species' responses depend on disturbance severity and are mediated by compensatory reproduction, immigration from refuge habitats, or both (Gido et al. 2019).

Native and nonnative species comprise the fish community of the upper Gila River. Four native species—longfin dace, *Agosia chrysogaster* (Leuciscidae); spikedace, *Meda fulgida* (Leuciscidae); loach minnow, *Rhinichthys* (*Tiaroga*) *cobitis* (Leuciscidae); and speckled dace, *Rhinichthys osculus* (Leuciscidae)—are small bodied with short life spans (maximum body length <150 mm, maximum age ≤3 years; table S.1; Olden et al. 2006). Three native species—roundtail chub, *Gila robusta* (Leuciscidae); Sonora sucker, *Catostomus insignis* (Catostomidae); and desert sucker, *Pantosteus clarkii* (Catostomidae)—are large bodied with longer life spans (maximum body length >300 mm, maximum age 13–15 years). Two nonnative species—yellow

bullhead, *Ameiurus natalis* (Ictaluridae), and smallmouth bass, *Micropterus dolomieu* (Centrarchidae)—are large bodied and long lived (7 and 14 years, respectively). Nonnatives were introduced in the early 1940s and were haphazardly stocked until the 1960s and 1970s (unpublished records from the New Mexico Department of Game and Fish [NMDGF]) approximately 13 and 11 generations ago, respectively. Prior to collection of tissue samples in 2010, populations of native species had not been artificially augmented or translocated.

Sixteen study sites (fig. 1) were selected on the basis of accessibility and availability of long-term monitoring data (five sites) and because they were representative of stream habitat types along the upper Gila River. The distribution of each species in the catchment varies such that not all species were present at all sites (fig. 1).

Demographic Data

Five sample locations coincided with NMDGF annual fish diversity survey stations (fig. 1), where the sampling protocol was designed to maintain similar effort (e.g., electrofishing time and number of seine hauls) at fixed locations (Propst et al. 2008). Presence-absence data from 1988 to 2010 were used to estimate annual extirpation probabilities (P_e). Detection histories for each species were fit to a multiple-season dynamic occupancy model (MacKenzie et al. 2003) using the program PRESENCE (Hines 2006) to estimate ε, the probability that an occupied site becomes unoccupied between years while accounting for imperfect detection. Annual monitoring consisted of single-pass surveys so detection probabilities (p) could not be simultaneously estimated. Instead, we used detection probabilities estimated by Whitney et al. (2016) using multipass surveys at a subset of locations that occurred from 2008 to 2011 (table S.1). A comprehensive occupancy analysis was beyond the scope of the present study and capabilities of the data set. We ran two models for each species, one in which we set all parameters, including ε and p, to be invariable across sites and years and a second in which we fixed p to values estimated by Whitney et al. (2016). Additionally, we estimated a naive ε that did not account for p to compare with model-based estimates. We converted estimates of ε to a per-generation P_e by taking the product of per annum ε and average age at maturity (a surrogate for generation time).

Census size estimates for each species were not available. Instead, we used a commonness index (C) to represent relative numerical differences across species. Using ecological survey data at all 16 sites collected between 2012 and 2017 (Whitney et al. 2016), we calculated relative abundance (percentage of total catch across sites) and percentage of occupied sites for each species. The commonness index C was calculated as the sum of relative abundance and percentage of occupied sites because both are percentage values. The index C incorporates both the number of individuals of each species and their spatial distribution; that is, widespread and abundant species have a high value of C, and rare species have a low value. We also calculated mean density of adults for each species at sites where adult individuals were present. We used length at maturation information compiled by Olden et al. (2006) to classify individuals as adults (table S.1).

Migration Rates and N_{ei}

We used a previously published data set of microsatellite genotypes (Pilger et al. 2017) to estimate contemporary gene flow (m, migration rate) and local genetic effective size for each species at each sample location. Genotypes were characterized at nine or ten microsatellite loci per species, and all loci conformed to neutral expectations (table S.2). Tissue samples for genetic assays were collected during fall 2010 from juveniles spawned earlier in the year, and care was taken to sample individuals of similar length (roughly between 50 and 75 mm in length). Thus, estimates of migration rate should apply to recently dispersed juveniles and their parents, while effective number of breeders and effective population size estimates should apply to the parents.

Contemporary estimates of migration (m) were obtained for each species using Bayesian inference according to the assignment model implemented in BIMr 1.0 (Faubet and Gaggiotti 2008). We used this model because it can be used with species exhibiting high gene flow and does not require subpopulations to conform to Hardy-Weinberg expectations. Reliable estimates of m require a minimum genetic differentiation between samples of $F_{ST} = 0.01$ (Faubet and Gaggiotti 2008). Because pairwise F_{ST} values among sample locations for some species were less than 0.01 and not significantly different from zero (Pilger et al. 2017), individuals across those locations were combined for analysis. No migration rate could be estimated for *A. natalis* because this species did not exhibit significant differentiation. See the supplemental PDF for details on which sample locations were combined for each species, BIMr model run settings, and output selection criteria. BIMr estimates unidirectional migration rates (m_{ij}, the rate from population j into i). We calculated the mean of off-diagonal m_{ij} estimates (i.e., when $i \neq j$) to estimate overall m for each species. Two independent approaches were used to quantify the level of asymmetric migration for each species. First, an index of asymmetry (A) was calculated from the unidirectional BIMr estimates using both the mean and the median of the absolute value of differences between m_{ij} and m_{ji}. Because the number of

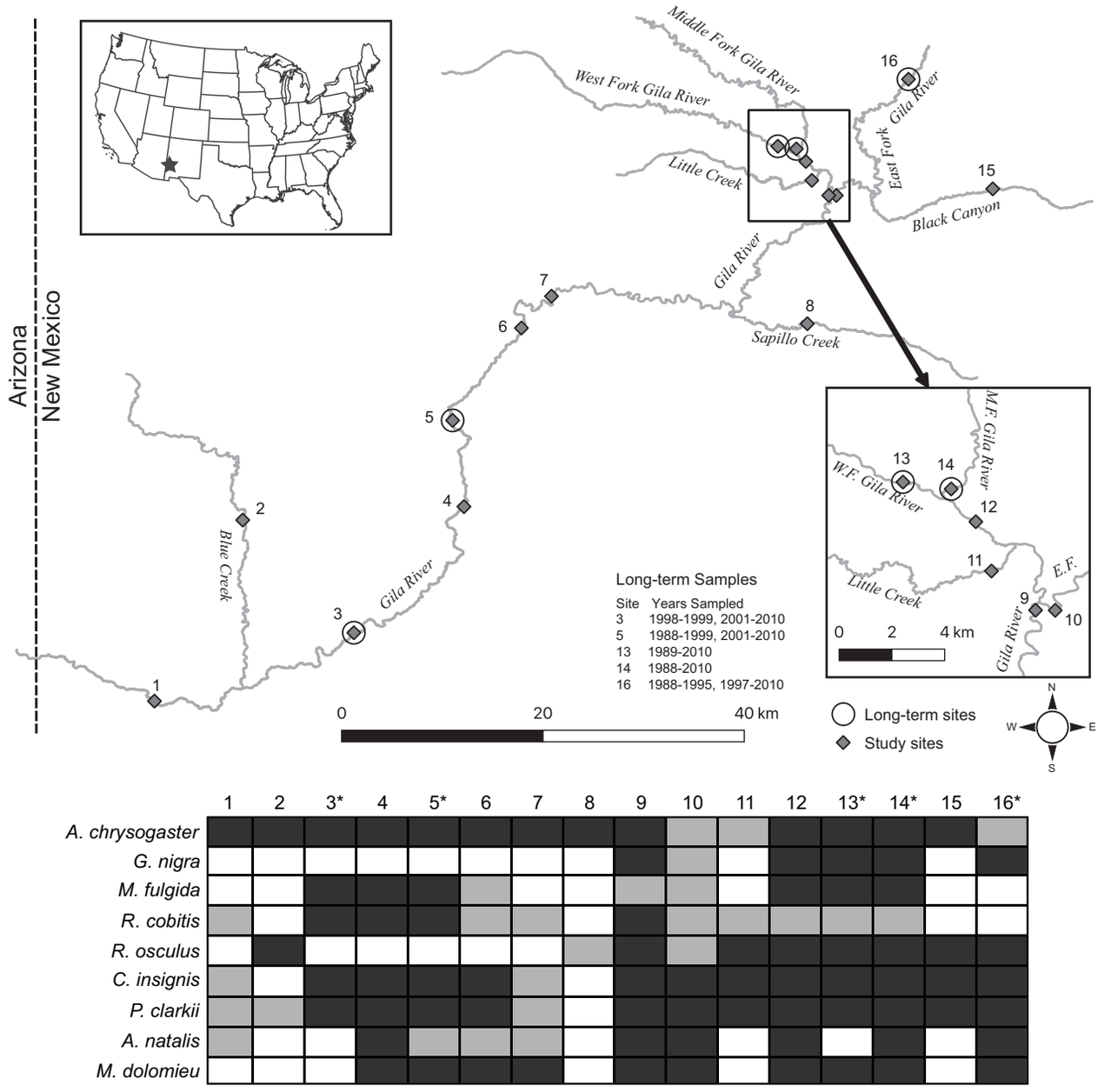

Figure 1: Sample locations where demographic and genetic data on nine fish species of the upper Gila River catchment were collected. The lower graphic depicts species' distributions across sites, with dark gray boxes indicating that a species was present at the time of sampling and light gray boxes indicating that a species has been documented at the location but was not present at the time of sampling. White boxes indicate that a species has not been documented at the location. Long-term monitoring sites are indicated with a circle (*top*) or an asterisk (*bottom*). Relative abundance and occupancy for calculating commonness as well as adult densities were collected from all 16 sites. Extirpation probabilities were estimated using data from long-term monitoring sites.

sites differed across species, the median value should minimize any effect this may have on estimates. Second, we calculated the coefficient of variation (CV) of allelic richness across sites (A_r, accounts for differences in sample size) because asymmetric dispersal of alleles should result in some subpopulations accumulating more alleles than others (Morrissey and de Kerckhove 2009). Note that the latter metric is a function of $N_e m$. Neither metric was calculated with respect to directionality, but both should inform the magnitude of asymmetry. Correlation

between these independently derived metrics of migration asymmetry was assessed using Pearson's correlation coefficient (r).

Standing levels of genetic diversity were quantified by estimating effective number of breeders (N_{bi}) and generational genetic effective size (N_{ei}) for each sampling location (i) for each species. Our sampling scheme involved collecting a random sample of young-of-year individuals. We used the sibship method (Wang 2009) implemented in COLONY (Jones and Wang 2010) and the linkage disequilibrium (LD) method (Hill 1981) implemented in LDNE (Waples and Do 2008), which when applied to our sampling scheme provided estimates of N_{bi}. Sample sizes within species varied across locations; therefore, we set a minimum sample size of 10 individuals per location for both methods. This resulted in exclusion of one location each for *A. natalis* and *G. robusta*, three locations for *C. insignis*, and four locations for *M. dolomieu*. Because of relatively small sample sizes at each location ($10 \leq n \leq 30$ individuals) we set the cutoff (P_{crit}) in LDNE to exclude low-frequency alleles at $P_{crit} = 1/2S$, where S is the sample size (Waples and Do 2010). In some instances, the LD method produced unrealistically large or infinite N_{bi} estimates with infinite upper confidence intervals (CIs). In these cases, we used the lower 95% CI as a plausible minimum effective size (Waples and Do 2010). Because of the sampling scheme, N_{ei} per generation cannot be estimated using the sibship method. Estimates of N_{bi} from the LD method were converted to N_{ei} using a two-trait approach described in Waples et al. (2014). Values for the two traits, adult life span and age at maturity (table S.1), were compiled by Olden et al. (2006). Total genetic diversity for each species was calculated by summing N_{bi} and N_{ei} across sample locations. Lower and upper 95% CIs were calculated as the sum of site-specific lower and upper CIs, respectively. Because the LD method produced infinite upper-bound CIs, the upper bound for $\sum N_{ei}$ was considered infinite if at least one site-specific upper bound was infinite.

Metapopulation Effective Size (metaN$_e$)

Metapopulation effective sizes for each species were calculated using N_{ei} estimates based on the corrected LD method and parameter values reported in table 3 according to the three models in table 2. For the pooled model ($metaN_{ePool}$), all individuals across sites were combined for a single N_e estimate following the LD method and two-trait correction for overlapping generations described above. We derived lower 95% CIs for the island model ($metaN_{eIsl}$; Wright 1931) by substituting the lower 95% CIs of $\sum N_{ei}$ and F_{ST} in the equation, recognizing the infinite upper bounds. For the extirpation model ($metaN_{eEx}$; Whitlock and Barton 1997), 95% CIs were derived from 1,000 simulated values by randomly drawing P_e from a binomial distribution with mean $n{\cdot}P_e$, keeping n fixed while drawing m and F_{ST} values assuming m was normally distributed (mean and standard deviation in table 3) and F_{ST} was uniformly distributed bounded by F_{ST} 95% CIs.

Statistical Analyses

We tested the hypothesis that standing levels of genetic diversity, estimated as $\sum N_{bi}$ and $\sum N_{ei}$, were associated

Table 3: Demographic and genetic parameter estimates used to calculate *metaN*$_e$ according to three models

Species (code)	n	N	Density (SD)	C	P_e	F_{ST}[a] (95% CI)	m (SD)	CV A_r[a]	A	CV N_{ei}
Leuciscidae:										
Agosia chrysogaster (Ac)	13	397	.022 (.035)	103.6	.06	.016 (.012 to .020)	.077 (.022)	.123	.088	.45
Gila robusta (Gr)	4	80	.0021 (.0022)	5.9	.83	.027 (.014 to .041)	.042 (.024)	.081	.047	.55
Meda fulgida (Mf)	6	170	.019 (.037)	37.4	.21	.021 (.015 to .026)	.057 (.031)	.14	.059	.47
Rhinichthys cobitis (Rc)	4	112	.012 (.015)	53.1	.06	.008 (.005 to .011)	.220 (.065)	.049	.196	.44
Rhinichthys osculus (Ro)	8	227	.015 (.019)	60.5	.14	.042 (.027 to .062)	.050 (.011)	.119	.005	.45
Catostomidae:										
Catostomus insignis (Ci)	9	292	.0063 (.0075)	82.3	.04	.014 (.010 to .020)	.102 (.032)	.113	.153	.79
Pantosteus clarkii (Pc)	12	344	.0087 (.0183)	88.6	.13	.013 (.009 to .016)	.070 (.026)	.096	.111	.86
Ictaluridae:										
Ameiurus natalis (An)	5	122	.0039 (.0046)	35.2	.23	.007 (−.004 to .024)	. . .	.02	. . .	.68
Centrarchidae:										
Micropterus dolomieu (Md)	6	195	.0037 (.0037)	15.4	.43	.036 (.024 to .048)	.114 (.067)	.037	.212	1.06

Note: Values include number of sample locations (n); total number of individuals sampled (N); mean density per square meter of adult-sized individuals with standard deviation; index of commonness (C), extirpation probability (P_e), global differentiation (F_{ST}), mean migration rate (m), and standard deviation (SD) of unidirectional migration rate estimates; coefficient of variation of allelic richness across locations (CV A_r); index of asymmetric migration (A); and CV of local N_es. Species codes are used in figures 2 and 3.

[a] Pilger et al. 2017.

with demographic patterns of commonness, mean adult densities, and P_e among species. We first used Spearman's rank-based measure of association to test for correlations among P_e, commonness, and density among species. Density ($\rho = -0.60$, $P = .097$) and commonness ($\rho = -0.83$, $P = .003$) were each correlated with P_e and with each other ($\rho = 0.70$, $P = .04$). Because of correlated variables, we used linear models to test the relationship of each explanatory variable on $\sum N_{bi}$ and $\sum N_{ei}$ separately. Commonness, adult density, $\sum N_{bi}$, and $\sum N_{ei}$ were transformed to natural log scale and P_e was transformed to log odds (i.e., logit) to better approximate normality of residuals. We considered relationships significant at $\alpha = .05$. To address questions related to intrinsic effects on $metaN_e$ and to facilitate comparisons among models, we calculated $metaN_e : \sum N_{ei}$ ratios for each metapopulation effective size model ($metaN_{ePool}$, $metaN_{eIsl}$, and $metaN_{eEx}$). We then calculated Pearson's correlation coefficients between $metaN_e$ ratios to metrics of migration asymmetry, global F_{ST}, and CV of N_{ei} to determine whether different $metaN_e$ models conformed to our predictions in table 2 based on assumptions of those models. All analyses were performed in R (R Core Team 2020). Data and scripts have been deposited in the Dryad Digital Repository (https://doi.org/10.5061/dryad.sxksn0332; Pilger 2022).

Results

Demographic Patterns

Mean adult density ranged from 0.002 per square meter for *Gila robusta* to 0.022 per square meter for *Agosia chrysogaster* (table 3). Small-bodied minnows exhibited higher adult densities than the large-bodied species. Commonness ranged from 5.9 for *G. robusta* to 103.6 for *A. chrysogaster*. The two sucker species were more common than all but one minnow species, *A. chrysogaster*. Estimates of ε were similar between model-based and naive estimates (results not shown) for seven species but diverged for *G. robusta* and *Meda fulgida* (table S.3). Per-generation P_e ranged from 0.04 for *Catostomus insignis* to 0.83 for *G. robusta*.

Migration Rates and N_{ei}

Migration rates were estimated for eight species because *Ameiurus natalis* lacked sufficient differentiation for the BIMr model. For *G. robusta* and *Micropterus dolomieu*, the model runs with the lowest deviance score showed evidence of nonconvergence, so we used estimates from a model run that had a higher deviance score but that had reached convergence. In all other species, the model run with the lowest deviance score also reached convergence. Unidirectional migration rates were variable across locations for each species, ranging from 0.0 to 0.48, indicating variable patterns of migration across species (figs. S.1–S.9). Mean values of m ranged from 0.042 (SD = 0.024) for *G. robusta* to 0.22 (SD = 0.065) for *Rhinichthys cobitis* (table 3). Estimates of A were similar using both mean (results not shown) and median absolute differences between m_{ij} and m_{ji}. The CV of A_r was negatively correlated with A (Pearson's $r = -0.73$, $P = .047$).

Site-specific estimates of N_{bi} based on the sibship method were consistently lower than estimates of N_{ei} using the LD method and had finite confidence intervals (table S.4). Estimates using the LD method frequently had infinitely large estimates or had infinitely large upper 95% confidence bounds. Of the 67 species-by-location combinations for which N_{ei} was estimated, 21 were finite (i.e., 95% CI not including infinity). Species with the largest proportion of finite estimates were *G. robusta* (0.75), *Pantosteus clarkii* (0.58), and *M. dolomieu* (0.50), whereas all other species only had one or two finite local N_{ei} estimates.

Demographic Associations with N_{bi} and N_{ei}

Standing levels of diversity ($\sum N_{bi}$ and $\sum N_{ei}$) were significantly related to demographic summary statistics (fig. 2). Both C and adult density explained greater than 60% of variation in $\sum N_{bi}$ and $\sum N_{ei}$. The sibship-based $\sum N_{bi}$ was positively related to C ($F_{1,7} = 16.0$, $P = .005$; adjusted $r^2 = 0.65$) and mean adult density ($F_{1,7} = 16.2$, $P = .005$; adjusted $r^2 = 0.66$) and was negatively related to P_e ($F_{1,7} = 8.3$, $P = .024$; adjusted $r^2 = 0.48$). From the LD method, $\sum N_{ei}$ was positively related to C ($F_{1,7} = 15.9$, $P = .005$; adjusted $r^2 = 0.65$) and mean adult density ($F_{1,7} = 15.6$, $P = .006$; adjusted $r^2 = 0.65$) and was negatively related to P_e ($F_{1,7} = 22.0$, $P = .002$; adjusted $r^2 = 0.72$).

Comparison of $metaN_e$ Models

The LD method frequently produced unbounded (no upper limit) estimates of N_{ei}, but pooling all individuals across samples resulted in bounded estimates for all species except *C. insignis* (table 4). As expected, $metaN_{ePool}$ was less than $\sum N_{ei}$ for all species, and the rank order of $metaN_{ePool}$ matched that of $\sum N_{ei}$; *C. insignis*, *A. chrysogaster*, and *P. clarkii* had the largest values of both metrics, and *A. natalis*, *G. robusta*, and *M. dolomieu* had the smallest values. The $metaN_{ePool} : \sum N_{ei}$ ratio was negatively related to F_{ST} ($F_{1,7} = 9.1$, $P = .020$; fig. 3).

For the island model, estimates of $metaN_{eIsl}$ were greater than estimates of $\sum N_{ei}$, with magnitudes explicitly related to F_{ST} as parameterized by the model (table 4). The $metaN_{eIsl} : \sum N_{ei}$ ratio showed no association with P_e, indices of gene flow asymmetry, or the CV of local N_{ei} (table 5).

Based on the extirpation model, the $metaN_{eEx} : \sum N_{ei}$ ratio ranged from 0.059 for *Rhinichthys osculus* to 0.64

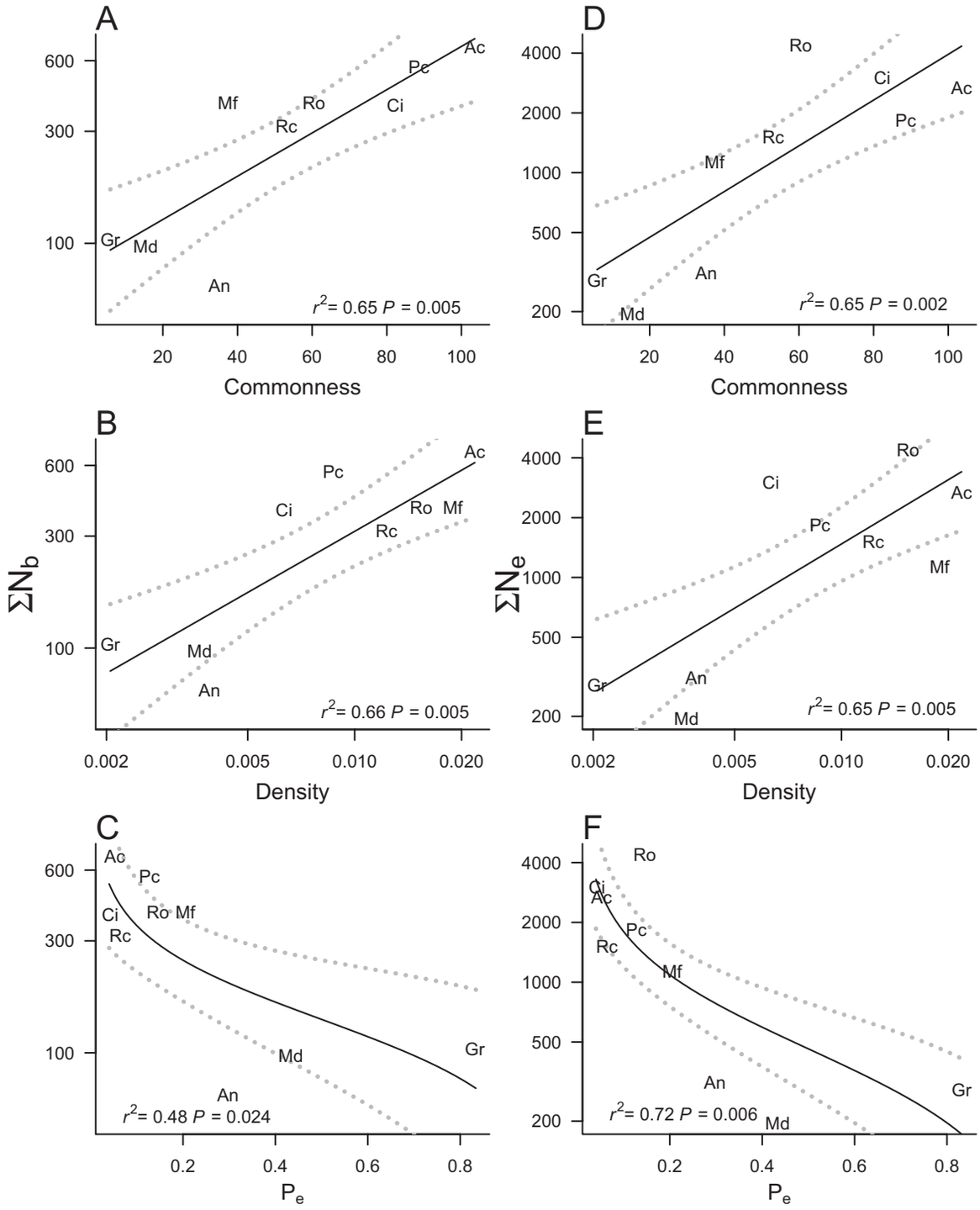

Figure 2: Relationship between demographic parameters and standing levels of genetic diversity, measured as effective number of breeders ($\sum N_{bi}$, *A*–*C*) and generational effective population size ($\sum N_{ei}$, *D*–*F*), for nine Gila River fish species. Demographic parameters are index of commonness (*A*, *D*), mean adult density (*B*, *E*), and P_e (*C*, *F*). Note that P_e was transformed to the log-odds scale for regression and back transformed for plotting. See table 3 for definitions of species codes. Note that the *y*-axes are on a log scale, as is density. Dotted lines show 95% confidence intervals.

Table 4: Standing levels of diversity and $metaN_e$ estimates according to three $metaN_e$ models

	Sibship $\sum N_{bi}$	LD $\sum N_{bi}$	$\sum N_{ei}$	$metaN_{ePool}$	$metaN_{eIsl}$	$metaN_{eEx}$	$metaN_{eEx}:\sum N_{ei}$
Agosia chrysogaster	686 (473–1,154)	1,940 (1,684–inf)	2,717 (2,358–inf)	1,657 (810–45,395)	2,761 (2,387–inf)	1,544 (683–5,236)	.568
Gila robusta	104 (58–213)	185 (92–inf)	209 (105–inf)	100 (73–152)	214 (106–inf)	42 (25–113)	.201
Meda fulgida	398 (228–5,489)	616 (578–inf)	1,271 (1,192–inf)	608 (456–878)	1,298 (1,210–inf)	271 (110–3,383)	.213
Rhinichthys cobitis	316 (186–1,198)	806 (569–inf)	1,662 (1,173–inf)	1,126 (546–38,730)	1,675 (1,179–inf)	441 (188–1,494)	.265
Rhinichthys osculus	396 (240–729)	1,088 (702–inf)	4,236 (2,734–inf)	475 (327–740)	4,422 (2,810–inf)	251 (91–1,482)	.059
Catostomus insignis	388 (236–689)	2,311 (1,081–inf)	2,825 (1,322–inf)	1,987 (743–inf)	2,865 (1,335–inf)	1,128 (472–3,493)	.399
Pantosteus clarkii	563 (344–1,014)	1,961 (720–inf)	1,781 (656–inf)	499 (332–912)	1,804 (661–inf)	1,144 (521–6,447)	.642
Ameiurus natalis	66 (33–158)	218 (16–inf)	377 (27–inf)	195 (71–4,261)	380 (27–inf)	...	...
Micropterus dolomieu	97 (54–207)	311 (69–inf)	368 (81–inf)	94 (48–231)	382 (83–inf)	76 (39–299)	.207

Note: $\sum N_{bi}$ was estimated using the sibship and linkage disequilibrium (LD) methods. $\sum N_{ei}$ and $metaN_{ePool}$ based on the LD method were corrected for overlapping generations according to Waples et al. (2014). $metaN_{eIsl}$ and $metaN_{eEx}$ were calculated using parameter estimates in table 3. $metaN_{eEx}$ could not be calculated for *A. natalis* because migration rates could not be estimated.

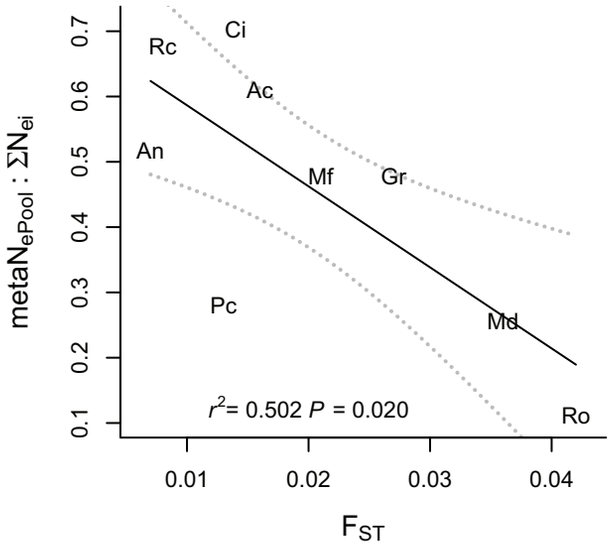

Figure 3: Ratio of $metaN_{ePool}$ to $\sum N_{ei}$ versus F_{ST} for nine species of fishes from the Gila River, New Mexico. See table 3 for definitions of species codes. Dotted lines show 95% confidence intervals.

for *P. clarkii* (table 4). The $metaN_{eEx}:\sum N_{ei}$ ratio showed no correlation with P_e, A, the CV of A_r, or the CV of local N_{ei}. This ratio was negatively associated with global F_{ST} across species, although marginally (table 5). Among the small-bodied minnows, *A. chrysogaster* had the highest $metaN_{eEx}:\sum N_{ei}$ ratio, followed by *R. cobitis* and *M. fulgida*, with *R. osculus* having the lowest. Except for *R. cobitis*, this order reflects degree of subdivision (based on F_{ST}) among these species (*A. chrysogaster* < *M. fulgida* < *R. osculus*).

Discussion

A network of permanent and semipermanent flowing waters comprises the upper Gila River catchment, providing the habitat template and disturbance regime for its fish community. By studying species within this shared abiotic backdrop, we were able to investigate how intrinsic traits and metapopulation dynamics influence genetic drift within (N_{ei}) and among ($metaN_e$) subpopulations over ecological timescales while controlling for extrinsic factors. Despite the dynamic nature of these metapopulations and an apparent lack of migration-drift equilibrium, standing levels of genetic diversity (measured as $\sum N_{bi}$ and $\sum N_{ei}$) conformed to predictions based on demographic patterns of commonness, adult densities, and extirpation probabilities across species. When empirical demographic and genetic data were substituted into $metaN_e$ models, results showed that population subdivision can depress $metaN_e$ relative to a panmictic population of the same size (decreasing $metaN_{eEx}:\sum N_{ei}$), consistent with genetic drift acting at the metapopulation scale. We argue that differences in traits and demography among species can either mitigate or exacerbate the impacts of genetic drift occurring in metapopulations that experience recurring extirpations.

Can Variation in Standing Levels of Genetic Diversity among Species Be Explained by Demographic Summary Statistics?

For Gila River fishes, patterns of abundance explained variation in genetic diversity, measured as $\sum N_{bi}$ and $\sum N_{ei}$ among species. Neutral genetic theory postulates that genetic diversity is positively related to abundance and its stability through time (Kimura 1983). Although qualified exceptions exist (e.g., Grundler et al. 2019; Rutherford et al. 2019), this relationship has been empirically confirmed across a broad taxonomic spectrum using a variety of putatively neutral loci (e.g., Frankham 1996; McCusker and Bentzen 2010; Hague and Routman 2016). We used two indices to characterize abundance patterns in the Gila River fish community, mean adult density and commonness, and observed positive relationships with genetic diversity among species. Except for *Agosia chrysogaster*, which had the greatest C and adult density of all species, the small-bodied, short-lived minnows (*Meda fulgida*, *Rhinichthys osculus*, and *Rhinichthys cobitis*) typically had higher densities of adults than the large-bodied, long-lived suckers (*Catostomus insignis* and *Pantosteus clarkii*), yet the suckers were more common than most minnows. Despite having lower densities of adults, the large-bodied, long-lived suckers had levels of genetic diversity comparable to those of minnows. Waples et al. (2013) found that delayed age at maturity increased N_e across a broad assortment of iteroparous taxa. For Gila River suckers and similarly long-lived organisms, high levels of diversity can be maintained with fewer numbers (relative to small-bodied species) of actively breeding adults from mixed cohorts. The species with the lowest diversity (native *Gila*

Table 5: Correlations between $metaN_e:\sum N_{ei}$ ratios and demographic and genetic summary statistics

	$metaN_{eIsl}$		$metaN_{eEx}$	
Variable	r	P	r	P
P_e	.36	.345	−.39	.336
F_{ST}	**1.0**	**<.001**	−.69	.059
A	−.38	.351	.21	.610
CV A_r	.21	.768	.16	.710
CV N_{ei}	.09	.825	.24	.572

Note: Pearson's correlation coefficients (r) were calculated across eight species excluding *Ameiurus natalis*, for which $metaN_{eEx}$ could not be estimated. $\sum N_{ei}$ was calculated on the basis of the linkage disequilibrium method corrected for overlapping generations. Boldface type indicates significance at α = .05.

robusta and nonnative *Ameiurus natalis* and *Micropterus dolomieu*) were not common, nor did they have dense aggregations of adults.

Our results provide evidence that recurring extirpations can have measurable impacts on genetic diversity. Mathematical models show that subpopulation extirpation and reformation acts as a secondary sampling process for genetic drift that can decrease neutral diversity (Slatkin 1977; Maruyama and Kimura 1980; Whitlock and Barton 1997; Pannell and Charlesworth 1999), yet empirical confirmation is scarce. Single-species investigations of invertebrate metapopulations have shown that repeated extirpations reduce genetic diversity (Ingvarsson et al. 1997; Walser and Haag 2012; Settepani et al. 2014). Comparison of genetic diversity among six midge species (genus *Chaoborus*) found ephemeral midges to have lower local diversity than permanent species, although this pattern was reversed at the regional scale (Berendonk et al. 2009). For Gila River fishes, increasing P_e was associated with lower $\sum N_{bi}$ and $\sum N_{ei}$. This relationship was primarily driven by *G. robusta*, *A. natalis*, and *M. dolomieu*. Low diversity could be explained by higher rates of population turnover but could be further depressed by other factors. For *G. robusta*, low $\sum N_{bi}$ and $\sum N_{ei}$ is likely the result of a persistent long-term decline due to negative interactions with introduced species and increasingly prolonged and severe droughts (Propst et al. 2008; Pilger et al. 2015). Nonnative *A. natalis* and *M. dolomieu* have persisted at relatively low abundance for approximately 13 and 11 generations, respectively, since cessation of sanctioned stocking. Whereas the number of generations past should be sufficient for any residual signal of anthropogenic-related linkage to dissipate (an artifact that can lower N_e with the LD method; Waples 2005), we cannot completely rule out their historical context.

Within their respective clades, patterns of $\sum N_{ei}$ versus P_e were consistent with predictions for both small-bodied minnows and larger suckers but less so with $\sum N_{bi}$, reflective of the fact that they estimate different aspects of a population. We were unable to further evaluate the relative impacts of adult density, C, and P_e on diversity because of the correlated nature of these variables, which are naturally confounded (Hanski and Gyllenberg 1997; Johnson 1998).

How Might Intrinsic Traits and Metapopulation Dynamics Influence metaN$_e$?

Basic understanding of how traits can influence genetic drift has primarily come from theoretical work. Slatkin (1977) modeled the genetic impacts of extirpations according to two types of founding group formation, propagule pool (colonists drawn from the same source subpopulation) and migrant pool (colonists drawn from all available subpopulations), revealing that characteristics of a recolonizing group had important consequences for genetic differentiation and diversity. Later, the continuum that these extremes represent was mathematically described in terms of k, the number of recolonizing individuals, and ϕ, the probability that recolonizing individuals are from the same extant subpopulation (Whitlock and McCauley 1990). Whitlock and Barton (1997) showed that the effect of extirpations for reducing *meta*N_e (i.e., *meta*$N_{eEx}:\sum N_{ei}$) was dependent on k and to a lesser extent ϕ, concluding that if k was small relative to local N_{ei} and ϕ approaches 1 (colonists from the same extant subpopulation), then extirpations should greatly reduce *meta*N_e. Yet even if k is large and $\phi = 0$ (colonists from all available extant subpopulations), then the effect of extirpations on *meta*N_e should be lessened but will never be greater than a panmictic population unless strict assumptions are met. Essentially, k and ϕ represent the patch recolonization component of genetic drift in metapopulations. These theoretical predictions suggest that ecological trade-offs resulting from different life history strategies may influence interspecific patterns in *meta*N_e.

For Gila River fishes, genetic drift appears to be a strong force for the three species with the lowest $\sum N_{ei}$. We could not directly estimate k and ϕ for Gila River fishes but rather inferred them by examination of intrinsic traits and demographic patterns. For example, *M. dolomieu*, and *A. natalis* grow to a larger size before maturity and trade off fecundity for high juvenile survivorship (e.g., nest guarding and territorial behavior; Winemiller and Rose 1992; table S.1). Although *G. robusta* may not exhibit the same trade-off, its persistent decline and low resilience to disturbances (Propst et al. 2008; Gido et al. 2019) suggest poor juvenile survival and recruitment. In these species, relatively few offspring combined with low abundance and few occupied sites suggests that the number of individuals (k) available to recolonize an unoccupied habitat is small with a high probability of being from the same extant subpopulation (ϕ approaching 1). This characterization along with high P_e suggests that genetic drift is a strong force in these species, congruent with low *meta*$N_e:\sum N_{ei}$ ratios.

Intrinsic traits and demography of other species suggest that genetic drift may be weaker across subpopulations. Large-bodied, long-lived suckers trade off juvenile survival for high fecundity (Winemiller and Rose 1992). Spawning coincides with snowmelt runoff dispersing large numbers of offspring downstream (Carter et al. 1986; Robinson et al. 1998). Offspring settle out in valley reaches where productivity is higher (Whitney et al. 2014). This life history strategy combined with high commonness ensures many colonists (k) with low probability of being from the same subpopulation (ϕ approaching 0), which should reduce the effects of genetic drift across subpopulations and result in higher *meta*$N_e:\sum N_{ei}$ ratios. Small-bodied, short-lived minnows (particularly *A. chrysogaster*, *M. fulgida*, and *R.*

osculus) have traits that allow them to use frequently disturbed habitats (Minckley and Deacon 1968; Minckley and Barber 1971; Whitney et al. 2016). Localized high densities of breeding adults and less passively dispersed offspring ensures many colonists (k) but with higher probability of being from geographically proximate sources than distant sources (moderate ϕ), resulting in moderate to high $metaN_e : \sum N_{ei}$.

What Can $metaN_e$ Models Reveal about the Effects of Genetic Drift Occurring at the Metapopulation Scale?

Whereas metapopulation effective size models may be restrictive and simplified, they can offer important insights into factors that exacerbate or mitigate genetic drift in natural populations. We used the pooled model to validate our assumption that $\sum N_{ei}$ is the expectation of a single panmictic population. Pooling samples across genetically differentiated populations can artificially increase the signal of linkage, causing the LD method to be biased downward (Waples and Do 2010). However, in a panmictic population with truly random mating, the linkage signal should be the same in individual subsamples as for pooled samples, resulting in $metaN_{ePool}$ equal to $\sum N_{ei}$ (i.e., $metaN_{ePool} : \sum N_{ei} = 1$). All species in this study exhibited some degree of differentiation and thus all had $metaN_{ePool}$ less than $\sum N_{ei}$, but the degree to which $metaN_{ePool}$ was lower was related to the level of differentiation across species. This pattern has been observed by others (Palstra and Ruzzante 2011; Turner et al. 2015). Because the observed pattern is most likely an artifact of the LD method, this finding alone is not sufficient to conclude that population subdivision reduces $metaN_e$.

Our data were least likely to conform to the island model (Wright 1943), yet none of the factors that violate its assumptions (increasing extirpation probability, variation of N_{ei}, or asymmetric migration across species) were correlated with $metaN_{eIsl} : \sum N_{ei}$ ratios. All species had $metaN_{eIsl} : \sum N_{ei}$ ratios greater than 1.0 because of the explicit parameterization using F_{ST}. Insensitivity of this model to violations limits the utility of this model to assess factors contributing to genetic drift for species that do not conform to its restrictive assumptions (e.g., Gomez-Uchida et al. 2013).

Although we predicted that violations of the extirpation model (Whitlock and Barton 1997) would manifest as negative correlations between $metaN_{eEx} : \sum N_{ei}$ ratios and increasing migration asymmetry and increasing variation in N_{ei}, these predictions were not confirmed. Whitlock and Barton (1997) derived the extirpation model to account for the component of genetic drift that occurs as a result of patch extirpation and recolonization. As the full model requires estimates of k and ϕ, we used the approximation that simplifies to the island model if extirpation probability is zero. As with the island model, the simplified parameterization of the extirpation model may not be informative for investigating the reduction in $metaN_e$ because of migration asymmetry or variable local effective sizes. Currently, few metapopulation models allow for local effective sizes to vary in space across subpopulations (but see Tufto and Hindar 2003). There was a marginal negative trend between $metaN_{eEx} : \sum N_{ei}$ and F_{ST}, indicating that for the levels of m and P_e observed here, population subdivision will tend to reduce $metaN_e$ in species experiencing extirpations (Whitlock and Barton 1997; Whitlock 2004). The $metaN_e$ models employed here allowed us to test predictions about how gene flow and genetic drift impact species with different intrinsic traits. Clearly, additional research and development of empirical approaches to estimate $metaN_e$—particularly methods that do not assume migration-drift equilibrium—are needed for more robust hypothesis testing.

Limitations of Analyses

N_{bi} *and* N_{ei}

Local estimates of N_{bi} and N_{ei} could be biased as a result of unsampled subpopulations or gene flow among subpopulations and from outside the system. The sibship method (Wang 2009) estimates an effective number of breeders from the number of full- and half-sibling pairs identified in a sample of genotypes and could underestimate the true N_{bi} if offspring from some parents were not sampled. This method is not influenced by gene flow or differentiation (Wang 2016). Gene flow among sampled and unsampled subpopulations within the upper Gila catchment could influence N_{ei} estimates from the LD method. Recent immigrants from highly differentiated sources can inflate the linkage signal, resulting in an underestimate of N_{ei}. Given that all species had overall low levels of differentiation (global $F_{ST} < 0.05$), any signal from recent immigrants is expected to be weak. Furthermore, the LD method is considered robust to gene flow (Waples and England 2011). Gene flow from outside the system is unlikely, as the Gila River mainstem downstream of the study area is more extensively modified and frequently dries (Turner and Propst 2016). The greatest source of uncertainty in local N_{ei} estimates is from the wide and often unbounded confidence intervals produced from the LD method. The lower-bound 95% confidence intervals provide a plausible estimate for the minimum value of N_{ei} (Waples and Do 2010); thus, estimates of $\sum N_{ei}$ are conservative.

P_e

Uncertainty in species' P_e arises from an inability to account for detection in the long-term diversity surveys and the limited number of locations with long-term data.

Estimates from single-pass surveys can be confounded with detection probability, resulting in estimates being biased to an unknown degree and direction (MacKenzie et al. 2003). We used the best available data on detection probabilities (Whitney et al. 2016) to minimize bias of these estimates. Our analyses assume that P_e estimated from five locations is representative for all subpopulations of each species. We feel this is a reasonable assumption because the ε values estimated using PRESENCE (Hines 2006) were consistent in rank order to those estimated by Whitney et al. (2016), who found no difference between naive and detection-corrected estimates.

Historical Context

Our estimates of N_{bi}, N_{ei}, and m reflect diversity and gene flow over the same temporal scale applicable to estimates of demographic parameters. Whereas others have documented that contemporary spatial distribution of genetic diversity can be explained by historical events (e.g., Salisbury et al. 2016; Ruzzante et al. 2019), historical contingency is less relevant to Gila River fishes. First, Pilger et al. (2017) found variable spatial patterns of diversity among these nine species that could be explained by combinations of landscape factors, intrinsic traits, and genetic drift. Second, the disturbance regime of this arid-land stream network results in frequent extirpation events (Propst et al. 2008; Gido et al. 2013, 2019). Recurring disturbances influence genetic diversity within and among subpopulations (e.g., Schrey et al. 2011; Davies et al. 2016) and lessen the influence of historical colonization patterns. Third, although the nonnative species were introduced within the last century, they never became abundant and widespread. Had either species increased in abundance, then inbreeding effective size (a measure of the amount of inbreeding in the generation prior to sampling; see Allendorf and Luikart 2009) would have increased, despite both having low absolute measures of diversity (e.g., number of alleles and heterozygosity; table S.2). Last, four of five minnow species exhibit similar intrinsic traits despite being evolutionarily divergent, and all five are part of a diverse lineage of minnows where life history traits associated with dispersal and persistence appear independently on the tree many times (Schönhuth et al. 2012).

Conclusion

Genetically effective population size, whether of a single population or a metapopulation, is an important metric for describing standing levels of genetic diversity and evolutionary trajectory (e.g., Frankham et al. 2014). Emergent properties and interactions arising from species' intrinsic traits, extrinsic factors, and metapopulation dynamics affect gene flow and genetic drift, which manifest as spatial patterns of genetic diversity and genetically effective metapopulation size. The variable yet largely unmodified nature of the upper Gila River catchment and its associated fish community provide a natural system to investigate how demography interacts with intrinsic traits to influence metapopulation effective size. This study provides empirical support for theory that predicts that population subdivision and extirpation-recolonization dynamics decrease metapopulation effective size but with a magnitude that depends on species-specific ecological characteristics (Whitlock and McCauley 1990; Whitlock and Barton 1997). For short-lived Gila River fish species, having high densities of breeding adults that rapidly colonize disturbed habitats is one strategy that maintains high standing levels of genetic diversity. For other species, genetic diversity can be maintained through age-structured populations of long-lived adults that can withstand years of poor juvenile recruitment. Considering how metapopulation dynamics impact genetic drift and gene flow is necessary to conserve each species' remaining genetic diversity at subpopulation and metapopulation scales. As the persistence of species is increasingly tied to demographic and genetic management, confirmation of these theoretical principles in a comparative context demonstrates how differences in species traits can have important implications for maintenance of genetic diversity. Last, our research highlights the need for comprehensive collection of demographic and genetic data to improve understanding of species-specific genetic patterns (e.g., Metcalf and Parvard 2007; Lowe and Allendorf 2010).

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Statement of Authorship

All authors were involved in the collection of data, contributed funding for field work, and edited versions of the manuscript. T.F.T. provided laboratory space and equipment to perform genetic assays. T.J.P. and T.F.T. were responsible for conceptual development of the manuscript. T.J.P. performed laboratory work and data analyses and wrote the manuscript.

Data and Code Availability

R code and data files to perform analyses have been deposited in the Dryad Digital Repository (https://doi.org/10.5061/dryad.sxksn0332; Pilger 2022).

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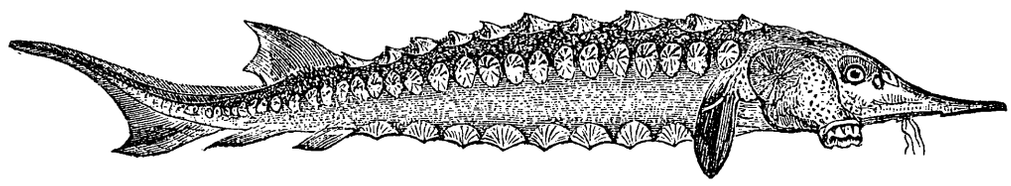

"The sturgeons . . . are genuine scavengers. With their long noses they turn up the bottoms of the streams and feed upon such organic materials they chance to find, using perhaps the flexible feelers beneath the snout to search out the exact location and nature of the food." Figured: "Sturgeon (*Acipenser oxyrhynchus*)." From "A Few Words about Scavengers" by Sanborn Tenney (*The American Naturalist*, 1877, 11:129–135).